Use the same corpus containing a total of 1000000 sentences that you developed in the previous assignment. Use the same training, development, and test splits. Develop unigram, bigram, trigram, and quadram language models corresponding to N=1,2,3,4 respectively under the following settings. Use the following formula for add-1/Laplace smoothing.

$$P_{Laplace}(w_n \mid w_1, w_2, \ldots, w_{n-1}) = \frac{C(w_1, w_2, \ldots, w_n) + 1}{C(w_1, w_2, \ldots, w_{n-1}) + V}$$

1. Unsmoothed - do not use any smoothing techniques, just use raw counts
2. Add one Smoothing
3. Add K smoothing (Choose K such that 0 < K < 1)
4. Interpolated Smoothing
5. Kneser-Ney Smoothing (with d=0.75)

For unigrams, use the following formula:

$$P_{\text{smooth}}(w_i) = \lambda P(w_i) + (1 - \lambda)\frac{1}{V})$$

Where λ is the interpolation constant and V is the vocabulary size.

Compare the perplexities of the language models.